

Company Fun Day & Game Night

Business Process Requirements - IBM BAW Training Exercise

Business Process	Company Fun Day & Game Night Request
Difficulty	Advanced
Main Roles	Employee, Department Manager, Finance, Event Operations, Safety, Department Director
Student Goal	Build the complete process and its user screens from this requirements document.

Build the solution exactly from the business rules below. The document describes expected business behaviour, not the technical implementation.

1. Process Overview

The company allows an employee to request a **Fun Day or Game Night**. The employee chooses the event details, food, games, prizes, and transportation. The request calculates an estimated budget and then goes through the required approvals and reviews before any booking starts.

The process must support approval, rejection, and rework. Some reviews and booking activities are conditional. Several reviews can run in parallel, and the process must wait for all required branches before it continues.

Process at a Glance

Stage	Business Flow
1	Employee creates and submits the Fun Day request.
2	Department Manager reviews the request.
3	Finance and Event Operations review in parallel. Safety joins the parallel stage only when required.
4	The process evaluates all required review results.
5	If the total cost is above 100,000, Department Director approval is required.
6	Required booking activities run in parallel.
7	After all required bookings finish, the employee receives the final event confirmation.

Main Business Rules

- Safety Review is required for an Outdoor event, more than 100 attendees, or when Mechanical Ride is selected.
- Director Approval is required when Total Estimated Cost is greater than 100,000.
- Food, activity, and transportation bookings are created only when the request needs them.
- The process can continue to final confirmation only after all required approvals and all required bookings are complete.
- If a completed review stage contains both Reject and Return for Modification decisions, Reject has priority and the request ends.

2. Business Information and Reusable Views

The solution uses the following business views. The same view can appear in different screens whenever that business information is needed.

View	Purpose
Event Details View	Main event information and attendee details.
Food View	Food selection and food cost.
Games & Activities View	Activities, competition information, and activity cost.
Transportation View	Transport request, bus calculation, and transport cost.
Budget Summary View	Combined event cost, service fee, total cost, and high-cost justification.
Approval View	Decision and comments used by reviewers and approvers.
Expected screen behaviour Automatically calculated values must be visible to the user but must not be changeable manually. When a rule says that an element is hidden, it must disappear from the screen until the business condition makes it relevant again.	

3. Process Steps and Screen Requirements

Step 1 - Create Fun Day Request

The Employee creates a new event request and chooses the services needed for the event.

Role	Employee
Views	Event Details View; Food View; Games & Activities View; Transportation View; Budget Summary View
User Actions	Enter and update the event information. Save the request or Submit it for approval.
Next Routing	Submit -> Manager Review. Save keeps the request available for later completion.
Visibility / Screen Behaviour	The views must react to the employee selections as defined in Section 4. Examples include Outdoor backup location, Food Required, Competition Mode, Transportation Required, and High-Cost Justification.

Step 2 - Manager Review

The Department Manager reviews the full request before it enters the specialist reviews.

Role	Department Manager
Views	Event Details View; Food View; Games & Activities View; Transportation View; Budget Summary View; Approval View
User Actions	Choose Approve, Reject, or Return for Modification. Enter comments when required.
Next Routing	Approve -> Parallel Review Stage. Reject -> Notify Employee -> End. Return for Modification -> Employee updates request -> Manager Review again.
Visibility / Screen Behaviour	All request information is shown using the current request data. The Approval View changes its comments behaviour based on the selected decision.

Step 3 - Parallel Review Stage

After Manager approval, the required specialist reviews begin in parallel. Finance and Event Operations are always required. Safety Review is included only when its business rule is true.

Step 3A - Finance Review

Role	Finance Officer
Views	Event Details View; Budget Summary View; Approval View
User Actions	Review the event cost. Choose Approve, Reject, or Return for Modification.
Next Routing	The decision waits for the other required parallel reviews before the process evaluates the combined result.
Visibility / Screen Behaviour	Budget values are shown using the latest request data. Approval comments follow the Approval View behaviour.

Step 3B - Event Operations Review

Role	Event Operations Officer
Views	Event Details View; Food View; Games & Activities View; Transportation View; Budget Summary View; Approval View
User Actions	Check whether the requested services can be arranged. Choose Approve, Reject, or Return for Modification.
Next Routing	The decision waits for the other required parallel reviews before the process evaluates the combined result.
Visibility / Screen Behaviour	The views show the current request and follow all visibility and behaviour rules defined in Section 4.

Step 3C - Safety Review

Role	Safety Officer
Views	Event Details View; Games & Activities View; Approval View
User Actions	Review the safety risk. Choose Approve, Reject, or Return for Modification.
Next Routing	This review exists only when Event Type = Outdoor, Number of Attendees > 100, or Mechanical Ride = Yes. Its result joins the other required reviews.
Visibility / Screen Behaviour	The screen is opened only for requests that require Safety Review. Approval comments follow the Approval View behaviour.

Step 4 - Evaluate Parallel Review Results

The process waits until all required specialist reviews finish. It then evaluates the combined decisions.

Role	System
Views	No user screen
User Actions	No user action.
Next Routing	If any required review = Reject -> Notify Employee -> End. Otherwise, if any required review = Return for Modification -> Employee updates request -> Manager Review again. If all required reviews = Approve -> check whether Director Approval is required.
Visibility / Screen Behaviour	Not applicable.

Step 5 - Director Approval

This step is required only for high-cost events.

Role	Department Director
Views	Event Details View; Budget Summary View; Approval View
User Actions	Choose Approve, Reject, or Return for Modification.
Next Routing	If Total Estimated Cost > 100,000: Director Approval is required. Approve -> Booking Stage. Reject -> Notify Employee -> End. Return for Modification -> Employee updates request -> Manager Review again. If Total Estimated Cost <= 100,000, this step is skipped and the process goes directly to Booking Stage.
Visibility / Screen Behaviour	The screen uses the latest event and budget information. Approval comments follow the Approval View behaviour.

Step 6 - Booking Stage

After all required approvals, the event services are booked. The required booking activities can run in parallel.

Role	Event Operations
Views	Food Booking: Event Details View + Food View. Activity Booking: Event Details View + Games & Activities View. Transportation Booking: Event Details View + Transportation View.
User Actions	Complete each required booking activity and confirm that the service is booked.
Next Routing	Food Booking runs only when Food Required = Yes. Activity Booking runs only when at least one activity is selected. Transportation Booking runs only when Transportation Required = Yes. The process waits for every required booking branch. If no booking branch is required, continue directly to Event Confirmation.
Visibility / Screen Behaviour	Only the business information needed for the current booking task is shown.

Step 7 - Event Confirmation

Role	Employee
Views	Event Details View; Food View; Games & Activities View; Transportation View; Budget Summary View
User Actions	Review the final event confirmation and complete the process.
Next Routing	Complete -> End.
Visibility / Screen Behaviour	The final screen shows the approved request and final calculated information.

Process Decision Summary

Decision	Rule
Safety Review	Required for Outdoor events, more than 100 attendees, or Mechanical Ride.
Director Approval	Required when Total Estimated Cost > 100,000.
Rework	Return for Modification sends the request back to the Employee and then through Manager Review again.
Booking	Only required service bookings are created; all required booking branches must finish before confirmation.

4. Detailed View Requirements

4.1 Event Details View

Element	Type	Business Requirement
Event Name	Text	Mandatory. Maximum 100 characters.
Event Type	Select	Mandatory. Allowed values: Indoor, Outdoor.
Event Date	Date	Mandatory. Must be a future date.
Number of Attendees	Integer	Mandatory. Must be greater than 0.
Duration	Decimal	Mandatory. Must be greater than 0 hours.
Location	Text	Mandatory. Maximum 150 characters.
Backup Location	Text	Required only for Outdoor events.
Estimated Staff Required	Integer	Automatically calculated. Visible but cannot be changed manually.

Event Details Behaviour

- When Event Type = Indoor, Backup Location is hidden.
- When Event Type = Outdoor, Backup Location is shown and becomes mandatory.
- Whenever Number of Attendees changes, Estimated Staff Required is recalculated as Number of Attendees / 25, always rounded up.
- Examples: 20 attendees -> 1 staff; 25 -> 1 staff; 26 -> 2 staff; 70 -> 3 staff.

4.2 Food View

Element	Type	Business Requirement
Food Required	Yes/No	Determines whether food is required.
Food Package	Select	Allowed values: Basic, Standard, Premium. Required when Food Required = Yes.
Price Per Person	Decimal	Automatically determined from Food Package. Visible but cannot be changed manually.
Vegetarian Meals	Integer	Must be 0 or greater. Cannot be greater than Number of Attendees.
Estimated Food Cost	Decimal	Automatically calculated. Visible but cannot be changed manually.

Food View Behaviour

- When Food Required = No, Food Package, Price Per Person, Vegetarian Meals, and Estimated Food Cost are hidden.
- When Food Required = Yes, the food details are shown.
- Basic Package -> Price Per Person = 150.
- Standard Package -> Price Per Person = 250.
- Premium Package -> Price Per Person = 400.
- Estimated Food Cost = Number of Attendees x Price Per Person.
- Whenever the food package changes, Price Per Person and Estimated Food Cost are recalculated.
- If Food Required changes from Yes to No: clear Food Package; set Price Per Person = 0; Vegetarian Meals = 0; Estimated Food Cost = 0.

4.3 Games & Activities View

Element	Type	Business Requirement
Board Games	Yes/No	Optional activity. Cost = 2,000 when selected.
Karaoke	Yes/No	Optional activity. Cost = 3,000 when selected.
Football Tournament	Yes/No	Optional activity. Cost = 5,000 when selected.
VR Tournament	Yes/No	Optional activity. Cost = 8,000 when selected.
Mechanical Ride	Yes/No	Optional activity. Cost = 15,000 when selected.
Competition Mode	Yes/No	Determines whether competition details are needed.
Number of Teams	Integer	Used only when Competition Mode = Yes. Must be greater than 1.
Prize Budget	Decimal	Used only when Competition Mode = Yes. Must be 0 or greater.
Activity Cost	Decimal	Automatically calculated from selected activities. Visible but cannot be changed manually.

Games & Activities Behaviour

- Activity Cost equals the total price of all selected activities.
- The Activity Cost must update whenever any activity selection changes.
- Example: Karaoke + VR Tournament -> 3,000 + 8,000 = 11,000.
- When Competition Mode = No, Number of Teams and Prize Budget are hidden.
- When Competition Mode = Yes, Number of Teams and Prize Budget are shown.
- If Competition Mode changes from Yes to No, clear Number of Teams and set Prize Budget = 0.

4.4 Transportation View

Element	Type	Business Requirement
Transportation Required	Yes/No	Determines whether transportation is required.
Pickup Location	Text	Required when Transportation Required = Yes.
Bus Capacity	Integer	Fixed value = 40. Visible but cannot be changed manually.
Number of Buses	Integer	Automatically calculated. Visible but cannot be changed manually.
Cost Per Bus	Decimal	Fixed value = 1,800. Visible but cannot be changed manually.
Estimated Transportation Cost	Decimal	Automatically calculated. Visible but cannot be changed manually.

Transportation Behaviour

- When Transportation Required = No, the transportation details are hidden.
- When Transportation Required changes to No, clear Pickup Location and set Number of Buses = 0 and Estimated Transportation Cost = 0.
- When Transportation Required = Yes, the transportation details are shown.
- Number of Buses = Number of Attendees / 40, always rounded up.
- Examples: 20 attendees -> 1 bus; 40 -> 1 bus; 41 -> 2 buses; 85 -> 3 buses.
- Estimated Transportation Cost = Number of Buses x Cost Per Bus.

4.5 Budget Summary View

Element	Type	Business Requirement
Food Cost	Decimal	Uses the current Estimated Food Cost.
Activity Cost	Decimal	Uses the current Activity Cost.
Prize Budget	Decimal	Uses the current Prize Budget.
Transportation Cost	Decimal	Uses the current Estimated Transportation Cost.
Subtotal	Decimal	Automatically calculated. Visible but cannot be changed manually.
Service Fee	Decimal	Automatically calculated. Visible but cannot be changed manually.
Total Estimated Cost	Decimal	Automatically calculated. Visible but cannot be changed manually.
High-Cost Justification	Text Area	Required only when Total Estimated Cost > 50,000.

Budget Calculation

- Subtotal = Food Cost + Activity Cost + Prize Budget + Transportation Cost.
- Service Fee = 10% of Subtotal.
- Total Estimated Cost = Subtotal + Service Fee.
- Whenever one of the cost values changes, Subtotal, Service Fee, and Total Estimated Cost must be recalculated.
- Example: Food 20,000 + Activities 11,000 + Prize 4,000 + Transportation 5,400 = Subtotal 40,400; Service Fee 4,040; Total 44,440.

High-Cost Behaviour

- When Total Estimated Cost > 50,000, High-Cost Justification is shown and is mandatory.
- When Total Estimated Cost <= 50,000, High-Cost Justification is hidden.
- If the request was previously above 50,000 and later becomes 50,000 or less, any previous High-Cost Justification is cleared.

4.6 Approval View

Element	Type	Business Requirement
Decision	Select	Allowed values: Approve, Reject, Return for Modification.
Comments	Text Area	Required for Reject and Return for Modification; optional for Approve.

Approval Behaviour

- When Decision = Approve, Comments are optional.
- When Decision = Reject, Comments are shown and mandatory.
- When Decision = Return for Modification, Comments are shown and mandatory.
- If the decision changes back to Approve, Comments are no longer mandatory.

5. Validation Rules

The request cannot be submitted when required information is missing or invalid.

- Event Name is mandatory and cannot exceed 100 characters.
- Event Date must be in the future.
- Number of Attendees must be greater than 0.
- Duration must be greater than 0.
- Location is mandatory.
- Outdoor events must have a Backup Location.
- If Food is required, a Food Package must be selected.
- Vegetarian Meals cannot be negative and cannot be greater than Number of Attendees.
- Competition Mode requires at least 2 teams.
- Prize Budget cannot be negative.
- Transportation requires a Pickup Location.
- High-Cost Justification is mandatory when Total Estimated Cost is greater than 50,000.
- Reject and Return for Modification decisions require Comments.

6. Routing and Decision Rules

Safety Review

- Required when Event Type = Outdoor OR Number of Attendees > 100 OR Mechanical Ride = Yes.
- If none of these conditions is true, the Safety Review branch is skipped.

Parallel Review Result

- The process waits for Finance, Event Operations, and Safety when Safety is required.
- If any completed required review is Reject, the request is rejected and the process ends.
- If there is no Reject but at least one required review is Return for Modification, the request goes back to the Employee and then returns to Manager Review.
- Only when all required reviews are Approve can the request move forward.

Director Approval

- Required when Total Estimated Cost > 100,000.
- Skipped when Total Estimated Cost <= 100,000.

Booking Stage

- Food Booking is required when Food Required = Yes.
- Activity Booking is required when at least one activity is selected.
- Transportation Booking is required when Transportation Required = Yes.
- All required booking branches may run in parallel.
- The process must wait for every required booking branch before Event Confirmation.

7. Expected Student Solution

The completed solution must include the full BPMN process and the required user screens.

- A process diagram that correctly handles approvals, parallel reviews, optional Safety Review, rework, conditional Director Approval, and conditional parallel bookings.
- Reusable views for Event Details, Food, Games & Activities, Transportation, Budget Summary, and Approval.
- Screens for the Employee request, Manager Review, Finance Review, Event Operations Review, Safety Review, Director Approval, booking activities, and final Event Confirmation.
- All field behaviour, calculations, visibility rules, data clearing rules, and validations described in this document.
- Correct routing for Approve, Reject, Return for Modification, conditional reviews, and conditional bookings.

End Condition

The event is confirmed only after all required approvals are completed and every required booking activity has finished.